

WAJAHAT AHMED

AI DEVELOPER | AI ENGINEER | GENERATIVE AI, COMPUTER VISION & MACHINE LEARNING

[Contact](#) [Email](#) [Website](#) [Google Scholar](#) [GitHub](#) [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Developer with 3+ years of industry experience designing, building, and deploying production AI systems across generative AI, computer vision, and machine learning. Experienced in RAG pipelines, multi-agent and agentic workflows, real-time video analytics, and scalable AI microservices using Python, PyTorch, LangChain/LangGraph, FastAPI, and Docker. Comfortable owning a solution end to end — from requirements gathering and architecture through deployment, monitoring, and customer support — for enterprise clients.

CORE SKILLS

- **Programming:** Python, C++, C, MATLAB, SQL
- **Generative AI & Agentic Systems:** LLMs, SLMs, RAG, LangChain, LangGraph, AI Agents, Multi-Agent Workflows, Prompt Engineering, Embedding Models, Vector Databases
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNNs (2D/3D), RNN/LSTM/GRU, Random Forest, XGBoost, LightGBM, Feature Engineering, Model Evaluation, Time-Series Analysis
- **Computer Vision:** OpenCV, YOLO, YOLO-Pose, MMDetection, MMPose, MediaPipe, Object Detection, Pose Estimation, Multi-Object Tracking, Image Classification, Video Analytics
- **AI Backend & MLOps:** FastAPI, REST APIs, Docker, CI/CD, Kafka, PostgreSQL, MongoDB, Redis, Milvus, Git, Linux
- **Cloud, Edge & Deployment:** Google Cloud Platform (GCP), local GPU infrastructure, Jetson Nano, Raspberry Pi, ESP32
- **Tools & Collaboration:** Jira, ClickUp, Notion, Slack, Agile

EXPERIENCE

AI Developer — Lambda Theta | Sep 2024 – Present

- Led the end-to-end design, development, and deployment of AI solutions for enterprise clients, owning solution architecture, technical decisions, production deployment, and customer delivery from requirements gathering to rollout.
- Owned the complete lifecycle of multiple AI features as the primary technical owner, covering architecture, implementation, deployment, maintenance, and customer support.
- Developed Retrieval-Augmented Generation (RAG) systems and agentic AI workflows using LangChain, LangGraph, embedding models, vector databases, and LLM APIs for enterprise document intelligence and autonomous multi-step reasoning.
- Architected scalable AI microservices with FastAPI, integrating PostgreSQL, Redis, and Milvus to support document ingestion, embedding generation, vector indexing, semantic retrieval, and AI-powered search.
- Designed and deployed production computer vision systems for real-time video analytics across distributed multi-camera environments, processing 20+ concurrent RTSP streams with low-latency inference.
- Built end-to-end machine learning pipelines covering data collection, annotation, preprocessing, model training, evaluation, deployment, monitoring, and continuous improvement on datasets exceeding 200K+ labeled samples and 1,200+ video sequences.
- Designed hybrid spatial-temporal deep learning architectures to improve recognition of complex human activities under viewpoint variation, occlusion, and concurrent actions.
- Evaluated and optimized classical and deep learning models — including Random Forest, XGBoost, LightGBM, MLP, RNN, GRU, and LSTM — to identify architectures balancing accuracy, latency, and computational efficiency in production.
- Fine-tuned and deployed deep learning models using PyTorch, YOLO-Pose, MediaPipe, and MMDetection for human activity recognition, drink spiking detection, and satellite imagery analysis.
- Investigated model failure modes, distribution shifts, and deployment challenges, applying geometric and statistical approaches to improve robustness and real-world generalization.
- Containerized AI services using Docker, integrated CI/CD pipelines, and supported scalable deployments across local GPU infrastructure and Google Cloud Platform (GCP).
- Collaborated directly with enterprise customers to translate business challenges into production-ready AI systems, delivering technical documentation, architecture designs, and project proposals.
- Mentored junior engineers and interns, providing technical guidance on AI system design, software engineering practices, and machine learning development.

Research Assistant — Neurocomputation Lab, National Center of AI | Sep 2022 – Sep 2024

- Designed and developed intelligent AIoT systems integrating machine learning, computer vision, embedded hardware, and real-time data acquisition for industrial monitoring and energy optimization.
- Built end-to-end machine learning workflows for large-scale sensor datasets, covering preprocessing, feature engineering, model training, evaluation, and statistical analysis.
- Evaluated and optimized machine learning models including Random Forest, Decision Tree, SVM, KNN, ANN, LSTM, and Hidden Markov Models to predict user-preferred fan speed while minimizing energy consumption.
- Deployed deep learning models on Raspberry Pi and embedded edge platforms for real-time computer vision inference under latency and computational constraints.
- Designed AIoT-enabled embedded hardware and low-power ESP32 firmware with BLE communication, heterogeneous sensor networks, and cloud connectivity for long-term field deployment.
- Designed experimental platforms and automated data acquisition pipelines for Li-ion battery characterization and health prediction.
- Conducted firmware security assessments for embedded controllers used in electric vehicle systems, identifying vulnerabilities and implementing mitigation strategies.
- Supervised undergraduate research in AI, computer vision, embedded systems, and robotics while authoring peer-reviewed publications, research proposals, and technical documentation.

SELECTED PROJECTS

- **Document Intelligence Platform (RAG-as-a-Service):** Architected a production-ready document intelligence platform supporting document ingestion, chunking, embedding generation, vector search, and semantic retrieval through modular AI microservices built with FastAPI, Kafka, and vector databases.
- **Multi-Agent Enterprise Assistant:** Built an agentic system in which coordinated LLM agents automate organizational workflows, combining tool use, role-based access control, and a dynamic agent registry for multi-step task execution.
- **Qiyas – Distributed Multi-Camera Activity Recognition:** Developed a scalable perception system for distributed multi-camera human activity recognition using hybrid spatial-temporal deep learning models, achieving 95% detection accuracy under real-world conditions.
- **Hazard Detection:** Developed a 3D CNN-based fire detection system for real-time CCTV surveillance, optimizing spatio-temporal feature extraction for early hazard recognition.
- **InstruX – Industrial Energy Intelligence Platform:** Developed an industrial AIoT platform integrating multiple energy analyzers for real-time monitoring, visualization, and predictive analytics, with machine learning models for energy forecasting and optimization; resulting journal paper under review.
- **SmartFan – AIoT Smart Fan System:** Designed an AIoT-enabled smart fan system integrating embedded hardware, IoT communication, and machine learning for energy optimization and user comfort, presented in a peer-reviewed publication.

EDUCATION & CERTIFICATION

- M.S. Artificial Intelligence — NED University of Engineering & Technology (2026 – Present)
- B.S. Applied Physics (Electronics Major, Gold Medalist) — NED University of Engineering & Technology (2018 – 2022)
- Google Cloud Certified — Associate Cloud Engineer (2026)

PUBLICATIONS

- Hashim Raza Khan, Wajahat Ahmed et al., "Design and Experimental Results of an AIoT-Enabled, Energy Efficient Ceiling Fan System," MDPI Sustainability, 2024.
- Two additional manuscripts currently under review (Results in Engineering; Scientific Data).